Work Integrated Learning Programmes Division
M.Tech (AIML)
Machine Learning

Assignment - 2

Marks: 15 Submission Deadline: 08-Nov-2026

1. Overview

In this assignment, you are required to:
- Implement multiple classification models
- Build an interactive Streamlit web application to demonstrate your models
- Deploy the app on Streamlit Community Cloud (FREE)
- Share clickable links for evaluation

You will learn real-world end-to-end ML deployment workflow: modeling, evaluation, UI design, and deployment.

The assignment has to be performed on BITS Virtual Lab. If you are still facing any issue with this, please send an email to csislabsupport@wilp.bits-pilani.ac.in and raj.rohan@wilp.bits-pilani.ac.in with subject as "NSP1 ML Assignment 2: BITS Lab issue" and get it resolved at the earliest.

2. Mandatory Submission Links

Each submission must be a single PDF file with the following (maintain the order):

1. GitHub Repository Link containing
    - Complete source code
    - requirements.txt
    - A clear README.md
    - Test data used in your experiments (csv)
2. Live Streamlit App Link
    - Deployed using Streamlit Community Cloud

    - Must open an interactive frontend when clicked
3. Screenshot
    - Upload screenshot of assignment execution on BITS Virtual Lab
4. The Github README content (details mentioned in Section 3 - Step 5) should also be part of the submitted PDF file.

3. Assignment Details

Step 1: Dataset choice

Choose ONE classification dataset of your choice from any public repository - Kaggle or UCI. It may be a binary classification problem or a multi-class classification problem.

Minimum Feature Size: 12
Minimum Instance Size: 500

Step 2: Machine Learning Classification models and Evaluation metrics

Implement the following classification models using the dataset chosen above. All the 5 ML models have to be implemented on the same dataset.

1. Logistic Regression
2. Decision Tree Classifier
3. K-Nearest Neighbor Classifier
4. Naive Bayes Classifier - Gaussian or Multinomial
5. Support Vector Machine

For each of the models above, calculate the following evaluation metrics:

1. Accuracy
2. AUC Score
3. Precision
4. Recall
5. F1 Score
6. Matthews Correlation Coefficient (MCC Score)

The assignment has to be performed on BITS Virtual Lab and a (ONE) screenshot has to be uploaded as a proof of that. [1 mark]

Step 3: Prepare Your GitHub Repository

Your repository must contain:

```
project-folder/
|-- app.py    (or streamlit_app.py)
|-- requirements.txt
|-- README.md
|-- test_data.csv
|-- model/ (saved trained models for all five classifiers, such as .joblib or .pkl files).
```

Include the source code (.py or .ipynb) and any required preprocessing artifacts.

Step 4: Create requirements.txt

Example:

streamlit
scikit-learn
NumPy
pandas
matplotlib
seaborn

Missing dependencies are the #1 cause of deployment failure, so be careful with this.

Step 5: README.md with the following structure. This README content should also be part of the submitted PDF file. Follow the required structure carefully.

a. Problem statement
b. Dataset description [1 mark]
c. Github Repository Link [1 mark for maintaining the github repo with all required files]
d. Models used: [5 marks - 1 marks for all the metrics for each model]
   Make a Comparison Table with the evaluation metrics calculated for all the 5 models as below:

ML Model Name	Accuracy	AUC	Precision	Recall	F1	MCC
Logistic Regression						

Decision Tree						
KNN						
Naive Bayes						
Support Vector Machine						

- Add your observations on the performance of each model on the chosen dataset. [3 marks]

ML Model Name	**Observation about model performance**
Logistic Regression	
Decision Tree	
KNN	
Naive Bayes	
Support Vector Machine	
Overall, Winner for your dataset?	

Step 6: Deploy on Streamlit Community Cloud

1. Go to https://streamlit.io/cloud
2. Sign in using GitHub account
3. Click “New App”
4. Select your repository
5. Choose branch (usually main)
6. Select app.py
7. Click Deploy

Your Streamlit app must include at least the following features : -

a. Dataset upload option (CSV) [As streamlit free tier has limited capacity, upload only test data] [1 mark]
b. Model selection dropdown (if multiple models) [1 mark]
c. Display of evaluation metrics [1 mark]
d. Confusion matrix or classification report [1 mark]

The uploaded test CSV must contain all required input features and the target column. The app must load the saved model, calculate the evaluation metrics, and display the results without retraining the model on the uploaded test data.

The results of different models on your "test data" should be visible on the streamlit app.

5. Anti-Plagiarism & Academic Integrity Guidelines

To ensure originality we will be performing the following checks. **Any plagiarism found will result in ZERO (0) marks.** Existing github repositories are also checked for plagiarism.

Code-Level Checks

- GitHub commit history will be reviewed
- Identical repo structure & variable names may be flagged

UI-Level Checks

- Copy-paste Streamlit templates without customization may be penalized

Model-Level Checks

- Same dataset + same model + same outputs across students will be investigated

Using AI tools is allowed only for learning support, not for direct copy-paste submissions.

8. Final Submission Checklist (Before You Submit)

☐ GitHub repo link works

☐ Streamlit app link opens correctly

☐ App loads without errors

☐ All required features implemented

☐ README.md updated and added in the submitted PDF.

The assignment is for **15 Marks** - Model implementation and uploading on GitHub (10 marks); Streamlit App Development (4 marks). Additional 1 mark is for performing the assignment on BITS Lab and uploading a screenshot of the same.

No extension of deadlines will be provided. Please submit within the deadline - 8 November 23:59 PM.

No DRAFT submissions will be accepted. Please remember to **SUBMIT** your assignment.

All the Best !!
Happy Learning 😳